

Dynamic Beam Clustering for Grid-of-Beam Multi-User Access in 6G LiFi Networks

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Abstract—The next generation of light fidelity (LiFi) networks will be based on laser-based optical wireless communication (OWC) to meet the demands for multi-Gigabit/s (Gb/s) per user data rates. In this paper, we consider a double-tier access point (AP) architecture using an array of arrays of vertical cavity surface emitting lasers (VCSELs) to provide grid-of-beam (GoB) multi-user access for sixth generation (6G) laser-based optical wireless networks. In particular, we develop a novel dynamic beam clustering (DBC) algorithm based on graph theory with the aim to perfectly cancel inter-cluster interference (ICI) by dynamically adapting the clustering strategy according to network variations. We then carry out performance analysis for key performance metrics including the multi-user sum rate and fairness, as well as energy efficiency (EE). Simulation results demonstrate remarkable performance gains of DBC with respect to static beam clustering (SBC) strategies using fixed cluster sizes especially in terms of the sum rate and EE.

Index Terms—Grid-of-beam (GoB), light-fidelity (LiFi), 6G, dynamic beam clustering (DBC), energy efficiency (EE).

I. INTRODUCTION

Data demands are surging as societies become increasingly data-driven. To meet these needs, millions of sensors will populate smart cities, homes, workplaces and production facilities, powered by artificial intelligence (AI)-driven systems [1]. Sixth generation (6G) communication networks will serve as the backbone, enabling massive high-speed data transfer to support emerging intelligent environments [1]. Beyond personalized communication, 6G will advance the Internet of Things (IoT) by interconnecting people, devices, sensors, wearables, and autonomous systems [2]. Indoor user equipment (UE) devices include smartphones, laptops, tablets, augmented reality (AR)/virtual reality (VR) glasses, gaming consoles with cloud streaming, and other smart devices such as home assistants, ultra-high-definition (UHD) televisions (TVs), and smart security systems, all driving unprecedented demand for ultra-high-speed 6G connectivity [2].

Optical wireless communication (OWC) technology, utilizing narrow infrared (IR) laser beams, has shown significant potential to deliver ultra-high-speed transmission in indoor multi-user wireless networks [3], [4]. Beam-steered IR light communication is tailored to attain multi-Gb/s data rates per UE by directing focused IR laser beams toward UEs [3]. As an alternative approach, Sun *et al.* have proposed a beam-domain massive multiple input multiple output (MIMO) OWC

network design by using a large transmit lens in front of a light-emitting diode (LED) array [5]. This lens refracts light from individual array elements in various directions, allowing the optical access point (AP) to communicate simultaneously with a large number of UEs over a wide coverage area. The findings in [5] indicate that, for a sufficiently large number of LEDs, the asymptotically optimal multi-user transmission policy under both total and per-LED power constraints is to employ non-overlapping light beams, a scheme referred to as beam division multiple access (BDMA).

Aiming to achieve ultra-high transmission capacities with full indoor coverage, we have explored laser-based IR OWC systems by proposing a novel access network design using vertical cavity surface emitting laser (VCSEL) arrays [6]. VCSELs are preferred over other types of laser diodes due to their outstanding features including high power efficiency, large modulation bandwidth, and well-controlled output beam characteristics, to name a few [7]. In [8], Liu *et al.* have proposed an optical wireless transmitter system design based on a VCSEL array with a multi-stage optics design to provide indoor multi-beam coverage with uniform beam spots. The authors have demonstrated a data rate of 8 Gb/s within a beam spot of $20 \times 20 \text{ cm}^2$ over a 3 m distance at a pre-forward error correction (FEC) bit error ratio (BER) of 3.8×10^{-3} .

When beams are massively deployed in an indoor network while partially overlapping for full beam coverage [6], a grid-of-beam (GoB) light fidelity (LiFi) network is formed [4]. A major challenge for multi-user LiFi network design based on a GoB approach is inter-beam interference (IBI), especially for UEs located in overlapping regions of multiple beams. Beam clustering in conjunction with coordinated multi-beam (CoMB)-joint transmission (JT) can effectively reduce inter-cluster interference (ICI) and hence improve the signal-to-interference-plus-noise ratio (SINR) performance. Inspired by coordinated multi-point (CoMP) clustering in radio frequency (RF) wireless networks [9], two schemes can be identified for beam clustering with CoMB-JT, namely static beam clustering (SBC) and dynamic beam clustering (DBC). In SBC, a predefined set of clusters with fixed sizes is devised, which remain unchanged despite any variations in the network. While SBC is a low-complexity scheme to improve the received SINR in each cluster, it does not efficiently use the system bandwidth

when the number of UEs is large, since it uses a fixed layout for clusters [4]. In this paper, we put forward a novel DBC scheme that completely eliminates ICI by adaptively changing the clustering strategy in response to interference patterns. We also conduct performance analysis in terms of multi-user sum rate and fairness, as well as energy efficiency (EE) to study the performance of the proposed DBC algorithm, corroborating its remarkable advantages compared to benchmark SBC schemes.

The rest of the paper is organized as follows. The GoB LiFi network architecture is expounded in Section II. The proposed DBC algorithm and the performance analysis are presented in Section III. Simulation results are discussed in Section IV. The key findings and future works are summarized in Section V.

II. GRID-OF-BEAM OPTICAL WIRELESS NETWORK

A. Array of Arrays AP Architecture

Fig. 1 depicts an indoor optical wireless network providing seamless GoB coverage based on an array of arrays of VCSELs. The proposed LiFi AP design is composed of 9 identical transmitter elements arranged as a 3×3 array with a 5×5 VCSEL array incorporated in each element, resulting in $N_{\text{VCSEL}} = 225$ in total. This architecture is also referred to as the double-tier AP in consideration of two-tier addressing required to reach each VCSEL in each array. Each transmitter element covers almost one ninth of the network area, as shown in Fig. 1.

Figs. 2(a) and 2(b) show the optical design of the double-tier AP. The adjacent transmitter elements are placed at a center-to-center distance of d_{lens} . In each array, the VCSELs are closely placed next to each other with a pitch distance of δ . Consequently, the beam spots may overlap significantly on the receiver plane. To separate the beams, a plano-convex lens can be placed in front of each VCSEL array. This lens refracts the light beams in different directions, enhancing spatial resolution on the receiver plane. The refraction angle for each beam depends on the lens thickness and the specific path it follows within the lens. To achieve the required spatial resolution, the lens parameters must be carefully adjusted according to the design specifications. The adjustable parameters of the plano-convex lens include the diameter L , the radius of curvature R_{lens} , and the center thickness τ_c [10].

B. Downlink System Model

Fig. 3 depicts the downlink system configuration where only the middle transmitter element of the double-tier AP is shown for convenience. A total of K UEs are uniformly distributed over the network area, with h_{DL} representing the vertical distance between the AP and the UEs. A single-mode VCSEL produces the fundamental transverse electromagnetic (TEM) mode, forming a spherical Gaussian beam with a circularly symmetric intensity profile, which is characterized by the beam waist radius w_0 and the wavelength λ [11]. For the k th UE

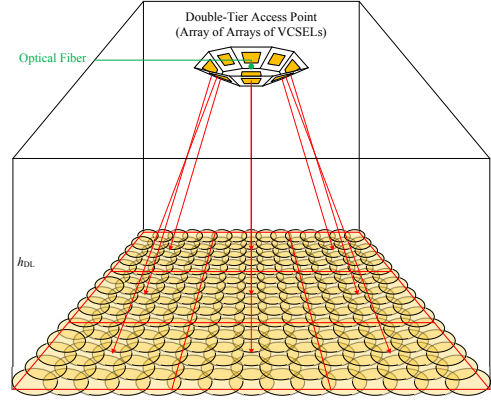


Fig. 1. Indoor GoB optical wireless access network using the double-tier AP design based on an array of arrays of VCSELs.

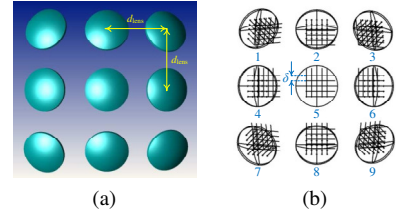


Fig. 2. LiFi AP architecture based on a 3×3 array of 5×5 VCSEL arrays: (a) Shaded top view, (b) Detailed top view.

located at coordinates $\mathbf{p}_k = [x_k \ y_k \ 0]^T$ on the receiver plane, the intensity profile of the i th beam can be expressed as:

$$I_i(x_k, y_k) = \frac{2P_t}{\pi w^2(z_{ik})} \exp\left(-\frac{2r_{ik}^2}{w^2(z_{ik})}\right), \quad (1)$$

where P_t is the transmit optical power, and $w(z)$ is the beam spot radius at distance z along the propagation axis, which is:

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2}, \quad z_R = \frac{\pi w_0^2}{\lambda}, \quad (2)$$

where z_R denotes the Rayleigh range. In (1), $r_{ik} = d_{kv} \sin \phi_{ik}$, $z_{ik} = d_{kv} \cos \phi_{ik}$, for $i = 1, 2, \dots, N_{\text{VCSEL}}$ and $v = \left\lceil \frac{i}{25} \right\rceil$, with $\lceil s \rceil$ denoting the smallest integer value satisfying $\lceil s \rceil \geq s$, and ϕ_{ik} is the radian angle of the i th beam relative to $\mathbf{p}_k$, which is calculated as:

$$\phi_{ik} = \cos^{-1} \left(\frac{\mathbf{d}_{kv} \cdot \mathbf{v}_i}{d_{kv}} \right). \quad (3)$$

In (3), $\mathbf{d}_{kv} = \mathbf{p}_k - \mathbf{q}_v$ represents the distance vector connecting the v th transmitter element with coordinates $\mathbf{q}_v = [x_v \ y_v \ z_v]^T$ to $\mathbf{p}_k$, $d_{kv} = \|\mathbf{d}_{kv}\| = \sqrt{(x_k - x_v)^2 + (y_k - y_v)^2 + z_v^2}$ is the corresponding Euclidean distance, $\mathbf{v}_i$ is the propagation vector of the i th beam, and $\cdot$ denotes the inner product.

In order to enable connectivity to multiple beams without strict alignment requirements, each UE is equipped with a non-imaging angle diversity receiver (ADR) based on compound parabolic concentrators (CPCs) [12]. The use of CPCs allows the maximum concentration gain to be realized due to the law of conservation of Etendue, and a multi-element ADR provides a robust performance over a wide field of view (FOV) [12].

We consider an ADR structure with seven receiver elements, each composed of a CPC coupled with a two-dimensional (2D) photodiode (PD) array. After performing equal gain combining (EGC) for each PD array, maximum ratio combining (MRC) is applied for the output signals of the receiver elements. The normal vector of the j th receiver element can be described by using its azimuth and elevation angles, denoted by ω_j and θ_j , respectively, as follows:

$$\mathbf{e}_j = [\sin \theta_j \cos \omega_j \quad \sin \theta_j \sin \omega_j \quad \cos \theta_j]^T, \quad (4)$$

where $\omega_j = (j-1)\frac{\pi}{3}$ and $\theta_j = 2\theta_{\text{CPC}}$ for $j = 1, 2, \dots, 6$, and $\omega_7 = \theta_7 = 0$ [12]. The direct current (DC) gain of the channel between the i th VCSEL and the j th receiver element of the k th UE can be derived as:

$$H_{kj}^{(i)} = \frac{2N_{\text{PD}}A_{\text{PD}}G_{\text{CPC}}}{\pi w^2(d_{kv} \cos \phi_{ki})} \exp\left(-\frac{2d_{kv}^2 \sin^2 \phi_{ki}}{w^2(d_{kv} \cos \phi_{ki})}\right) \times \cos \psi_{ji} \mathbb{1}_{\theta_{\text{CPC}}}(\psi_{ji}), \quad (5)$$

where A_{PD} and N_{PD} denote the PD area and the number of PDs on the array, respectively, $G_{\text{CPC}} = \frac{n_{\text{CPC}}^2}{\sin^2 \theta_{\text{CPC}}}$ represents the CPC gain for a refractive index of n_{CPC} and an acceptance angle of θ_{CPC} , and $\psi_{ji} = \cos^{-1}(\mathbf{e}_j \cdot \mathbf{v}_i)$ is the incidence angle of the i th beam on the j th receiver element. The last factor in (5) is an indicator function that is defined as $\mathbb{1}_{\theta_{\text{CPC}}}(\psi_{ji}) = 1$ if $0 \leq \psi_{ji} \leq \theta_{\text{CPC}}$, and 0 otherwise. Note that the half-angle FOV of the ADR is $\text{FOV} = 3\theta_{\text{CPC}}$ [12], as shown in Fig. 3.

C. Coordinated Multi-Beam Transmission

When multiple beams share the same frequency spectrum, there is IBI. CoMB-JT is used to reduce IBI by coordinating transmissions across a cluster of beams. To this end, the coverage area of the network is divided into N_c distinct regions, each served by a dedicated cluster. Each UE is supported by one cluster within a region determined by the beam clustering scheme. Coordinated VCSELs send multiple replicas of the same signal for each UE. Based on intensity modulation and direct detection (IM/DD), these signals are detected simultaneously at the receiver, allowing their received powers to be combined, thereby improving the received SINR.

To elaborate, let $\mathcal{C}_u$ denote the index set of the VCSELs involved in the u th cluster, for $u = 1, 2, \dots, N_c$, which serves a subset of the UEs with K_u members. The UEs in the network are globally indexed from 1 to K and those served by the u th cluster are indicated by the index set $\mathcal{K}_u$ with cardinality $|\mathcal{K}_u| = K_u$. The equivalent DC gain of the CoMB-JT channel between the u th cluster and the k th UE is given by:

$$H_k^{\mathcal{C}_u} = \sum_{j=1}^7 H_{kj}^{\mathcal{C}_u} = \sum_{j=1}^7 \sum_{i \in \mathcal{C}_u} H_{kj}^{(i)}, \quad (6)$$

where $H_{kj}^{\mathcal{C}_u} = \sum_{i \in \mathcal{C}_u} H_{kj}^{(i)}$ is the corresponding channel gain for the j th receiver element.

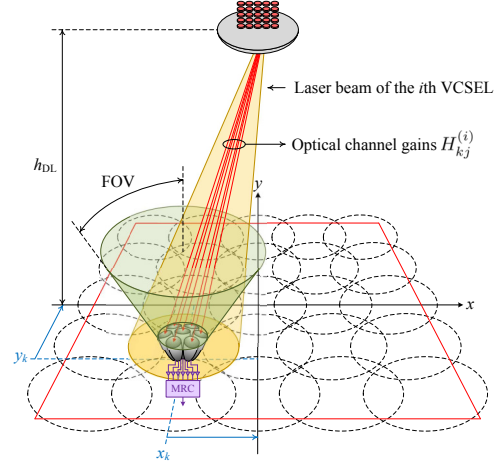


Fig. 3. Downlink system model representing the optical channel gains $H_{kj}^{(i)}$ from the i th VCSEL to the j th element of the ADR, $j = 1, 2, \dots, 7$, for the k th UE. Note that only the central element of the AP is shown.

III. DYNAMIC BEAM CLUSTERING

A. Proposed Algorithm

In SBC, a predefined set of clusters of fixed sizes is devised, which remain unchanged despite any network variations. By contrast, we propose DBC to dynamically adjust the clustering strategy in response to variable network loads due to an uneven distribution of UEs. The execution of DBC encompasses two steps including UE-beam association and cluster formation. At first, each UE needs to be associated with at least one beam, as it may be located within the beam spot of one or more beams depending on its position on the receiver plane. When a UE is located in the overlapping region of multiple beams, it is associated with all of them. This procedure leads to a $K \times N_{\text{VCSEL}}$ binary-valued matrix, denoted by $\mathbf{M} = [m_{ij}]$.

In DBC, for a given distribution of UEs, clusters are formed such that ICI is perfectly canceled. The main criterion used for this purpose is to cluster any two beams if there is at least one UE in their overlapping region. Hence, even if there are UEs that lie in different clusters while their clusters are physically overlapping, they receive ICI-free signals by definition. Note that the cluster size can be as small as one, indicating that there is a single beam with no UE in common with its adjacent beams. To develop an algorithm for DBC as described, we apply graph theory. To this end, we model the beams and the UEs as nodes and edges of the graph, respectively. Using the computed UE-beam association matrix, the adjacency matrix of the graph, denoted by $\mathbf{A} = [a_{ij}]$, which is a binary-valued matrix of size $N_{\text{VCSEL}} \times N_{\text{VCSEL}}$, is obtained by assigning 1 (0) to a_{ij} if at least one UE (no UE) is shared by beam i and beam j . The resulting graph is a disconnected graph with multiple connected subgraphs. Each subgraph represents a distinct cluster and its nodes are the desired index set of the beams involved in that cluster. The proposed DBC algorithm is summarized in four steps in Algorithm 1. In line 9, the breadth-first search (BFS) [13] is used to find the nodes of

each subgraph. Note that the index set of beams for the clusters of size one is separately computed in Step 3.

B. Sum Rate Analysis

To effectively eliminate intra-cluster multi-user interference (MUI), we use orthogonal frequency division multiple access (OFDMA) based on DC-biased optical OFDM (DCO-OFDM). When using OFDMA, a fraction $b_k^{C_u}$ of the total bandwidth and a fraction $p_k^{C_u}$ of the total power are allocated to the k th UE in the u th cluster subject to $\sum_{k=1}^{K_u} b_k^{C_u} = 1$ and $\sum_{k=1}^{K_u} p_k^{C_u} = 1$. Based on MRC, the combining weight for each ADR element is proportional to its received photocurrent to noise ratio [14]. Therefore, since ICI is canceled by DBC, the resulting signal-to-noise ratio (SNR) for the k th UE can be derived as:

$$\text{SNR}_k^{C_u} = \sum_{j=1}^7 \text{SNR}_{kj}^{C_u} = \sum_{j=1}^7 \frac{p_k^{C_u} P_{\text{elec}} \left(\eta_{\text{PD}} H_{kj}^{C_u} \right)^2}{b_k^{C_u} \xi^2 \sigma_{kj}^2}, \quad (7)$$

where $\text{SNR}_{kj}^{C_u}$ is the corresponding SNR for the j th receiver element, $\xi = \frac{N-2}{N}$ is the subcarrier utilization ratio using N subcarriers, η_{PD} is the PD responsivity, P_{elec} is the average electrical power which is related to the average transmit optical power, denoted by P_t , via $P_t = k_{\text{DC}} \sqrt{P_{\text{elec}}}$, where k_{DC} is the DC bias scaling factor [15]. A value of $k_{\text{DC}} = 3$ is commonly used to ensure that the clipping distortion remains negligible. The total noise detected by the j th receiver element of the k th UE is modeled as a white Gaussian noise with a single-sided power spectral density (PSD) of:

$$S_{kj} = \frac{4\kappa T}{R_L} F_n N_{\text{PD}} + 2q \sum_{u=1}^{N_c} \eta_{\text{PD}} P_{kj}^{C_u} + \text{RIN} \sum_{u=1}^{N_c} \left(\eta_{\text{PD}} P_{kj}^{C_u} \right)^2, \quad (8)$$

where κ is the Boltzmann constant, T is temperature in Kelvin, R_L is the load resistance, F_n is the preamplifier noise figure, q is the elementary charge, RIN is the PSD of the relative intensity noise (RIN), and $P_{kj}^{C_u} = H_{kj}^{C_u} P_t$. The total noise variance is $\sigma_{kj}^2 = S_{kj} B$, where B is the single-sided bandwidth of the system.

By using adaptive quadrature amplitude modulation (QAM), the achievable rate for the k th UE is given by [12]:

$$R_k^{C_u} = b_k^{C_u} \xi B \log_2 \left(1 + \frac{\text{SNR}_k^{C_u}}{\Gamma} \right), \quad \Gamma = -\frac{\ln(5 \text{ BER})}{1.5}, \quad (9)$$

where Γ is the SNR gap due to the required BER performance. The sum rate for all UEs is computed by:

$$R = \sum_{u=1}^{N_c} \sum_{k=1}^{K_u} R_k^{C_u} = \xi B \sum_{u=1}^{N_c} \sum_{k=1}^{K_u} b_k^{C_u} \log_2 \left(1 + \frac{\text{SNR}_k^{C_u}}{\Gamma} \right). \quad (10)$$

The multi-user fairness is measured by means of Jain's fairness index [16], which is given by:

$$J = \frac{\left(\sum_{u=1}^{N_c} \sum_{k=1}^{K_u} R_k^{C_u} \right)^2}{K \sum_{u=1}^{N_c} \sum_{k=1}^{K_u} \left(R_k^{C_u} \right)^2}. \quad (11)$$

Algorithm 1 Dynamic Beam Clustering

Input: $K \times N_{\text{VCSEL}}$ UE-beam association matrix, $\mathbf{M}$

Output: Index sets of beams for all clusters $\mathcal{C}$, total number of clusters N_c , and number of beams for each cluster $N_b^{C_u}$ for $u = 1, 2, \dots, N_c$

Step 1 – Compute the adjacency matrix $\mathbf{A}$

```

1:  $\mathcal{U}_2 \leftarrow$  The index set of the UEs associated with at least two beams
2:  $U_2 \leftarrow |\mathcal{U}_2|$ 
3:  $\mathbf{A} \leftarrow \mathbf{0}$ 
4: for  $k = 1, 2, \dots, U_2$  do
5:    $\mathcal{B}_k \leftarrow$  The index set of overlapping beams for the  $k$ th entry of  $\mathcal{U}_2$ 
6:    $\mathbf{A}(\mathcal{B}_k, \mathcal{B}_k) \leftarrow 1$ 
7: end for

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Step 2 – Find the index sets of clustered beams

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8: Define  $\mathcal{S}$  as the set of nodes for all the subgraphs
9:  $\mathcal{S} \leftarrow \text{BFS}(\mathbf{A})$ 
10:  $N_c \leftarrow |\mathcal{S}|$ 
11:  $\mathcal{C} \leftarrow \phi$ 
12: for  $u = 1, 2, \dots, N_c$  do
13:    $\mathcal{C}_u \leftarrow$  The set of nodes for the  $u$ th subgraph,  $\mathcal{S}_u \subset \mathcal{S}$ 
14:    $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}_u$ 
15:    $N_b^{C_u} \leftarrow |\mathcal{C}_u|$ 
16: end for

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Step 3 – Find the index set of beams constituting single-beam clusters

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17:  $\mathcal{U}_1 \leftarrow$  The index set of the UEs associated with only one beam
18:  $U_1 \leftarrow |\mathcal{U}_1|$ 
19: for  $k = 1, 2, \dots, U_1$  do
20:    $b_k \leftarrow$  The beam index for the  $k$ th element of  $\mathcal{U}_1$ 
21:   if  $b_k \notin \mathcal{S}$  then
22:      $N_c \leftarrow N_c + 1$ 
23:      $u \leftarrow N_c$ 
24:      $\mathcal{C}_u \leftarrow \{b_k\}$ 
25:      $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}_u$ 
26:      $N_b^{C_u} \leftarrow 1$ 
27:   end if
28: end for

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Step 4 – Return the output

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29: return  $\mathcal{C}$ ,  $N_c$  and  $N_b^{C_u}$  for  $u = 1, 2, \dots, N_c$ 

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C. Energy Efficiency Analysis

The EE of a wireless network is defined as the number of bits that can be reliably transmitted per unit of energy consumed [17]. It can be considered as a quality of service (QoS)-to-cost ratio to assess the bit delivery efficiency of the network. For the GoB optical wireless network, the EE is evaluated as follows:

$$\text{EE} = \frac{R}{\text{PC}}, \quad (12)$$

where R is the multi-user sum rate given by (10), and PC is the total power consumption (PC) of the double-tier AP with beam clustering. In DBC, a variable number of N_b beams are used depending on the number of clusters formed and the number of beams involved in each cluster. As a result, $N_b = \sum_{u=1}^{N_c} N_b^{C_u}$, where $N_b^{C_u}$ indicates the number of beams that participate in the u th cluster, which is computed by the DBC algorithm. Consequently, the total PC is $\text{PC} = N_b \text{PC}_{\text{Tx}}$, where PC_{Tx} is the transmitter PC per VCSEL. With a digital implementation of DCO-OFDM based on inverse fast Fourier transform (IFFT), the transmitter involves an N -point IFFT

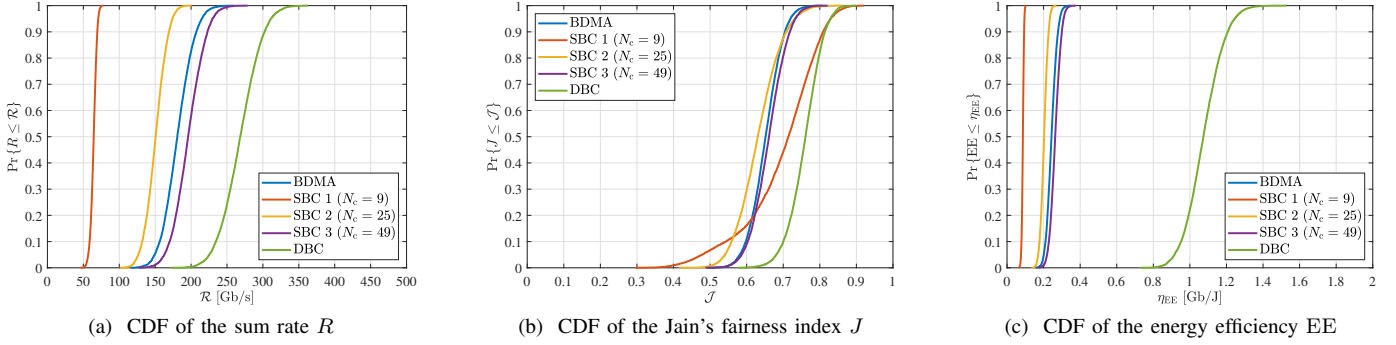


Fig. 4. Statistical performance of DBC for $K = 50$.

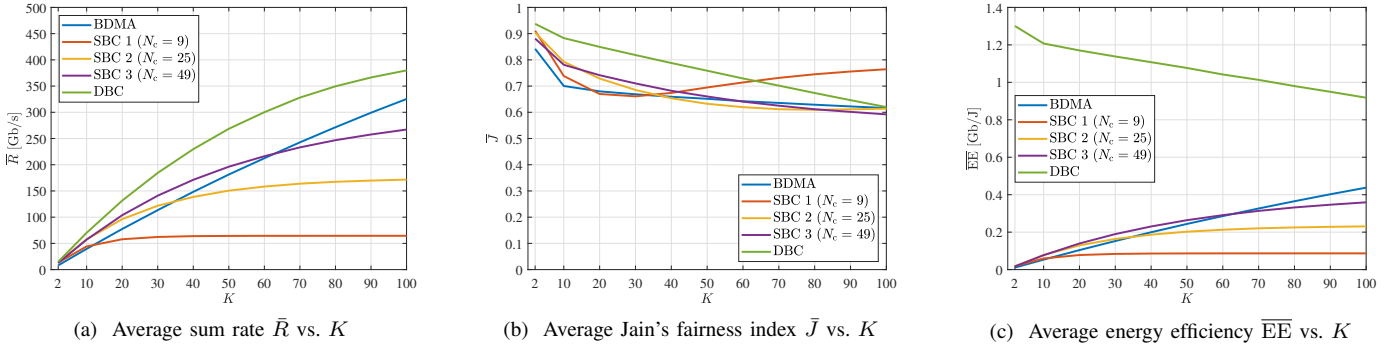


Fig. 5. Average performance of DBC vs. Number of UEs K .

operation, digital-to-analog conversion (DAC), DC biasing and amplitude scaling of the modulated signal. In other words:

$$PC_{Tx} = PC_{IFFT} + PC_{DAC} + PC_{DC} + PC_{AC}, \quad (13)$$

where PC_{DC} is the DC power and PC_{AC} is the alternating current (AC) power of the DCO-OFDM signal. According to [18], [19], the first term in (13) can be expressed as:

$$PC_{IFFT} = \left(\frac{1 + \eta_{leak}}{\eta_{CPE}} \right) (8 \log_2 N) B, \quad (14)$$

where η_{leak} accounts for the power leakage in complementary metal-oxide semiconductor (CMOS) switches, and η_{CPE} is the computational power efficiency (CPE) in giga operations per second (GOPS) per watt (GOPS/W). For the last two terms in (13), provided that the VCSEL operates in its linear dynamic range, it can be shown that:

$$PC_{AC} + PC_{DC} = R_{VCSEL} \left(\frac{1 + k_{DC}^2}{k_{DC}^2} \right) \left(\frac{P_t}{\eta_{VCSEL}} \right)^2, \quad (15)$$

where R_{VCSEL} and η_{VCSEL} are the equivalent load resistance and the electrical-to-optical conversion efficiency, respectively, from the VCSEL's light-current-voltage (LIV) characteristics.

IV. NUMERICAL RESULTS

In this section, we evaluate the performance of the proposed DBC for a GoB multi-user LiFi network based on the double-tier AP. The simulation setup consists of an indoor environment as depicted in Fig. 1 with dimensions of $5 \times 5 \times 3$ m³. Simulation parameters are listed in Table I. For EE evaluation:

TABLE I
SIMULATION PARAMETERS

Parameter	Description	Value
h_{DL}	Vertical separation	3 m
L	Lens diameter	16 mm
R_{lens}	Radius of curvature	15 mm
d_{lens}	Lens-to-lens distance	2 cm
λ	VCSEL wavelength	940 nm
P_t	Transmit power per VCSEL	10 mW
RIN	RIN PSD	-155 dB/Hz
n_{CPC}	CPC refractive index	1.7
η_{PD}	PD responsivity	0.6 A/W
N_{PD}	Number of PDs per PD array	16
FOV	ADR half-angle FOV	45°
F_n	Preamplifier noise figure	5 dB
B	System bandwidth	2 GHz
N	IFFT/FFT size	1024
BER	Pre-FEC BER	3.8×10^{-3}

$\eta_{leak} = 0.9$ [20], $\eta_{CPE} = 850$ GOPS/W [21], $R_{VCSEL} = 50 \Omega$ and $\eta_{VCSEL} = 0.5$ W/A. As the benchmark for comparison, we consider four SBC schemes. The baseline SBC scheme is referred to as BDMA in which the bandwidth is fully reused across all the beams, providing the highest possible spatial reuse of the spectral resources and at the same time causing the highest level of interference. The other three schemes are referred to as SBC 1, SBC 2 and SBC 3 with a fixed number of clusters of $N_c = 9, 25, 49$, respectively. SBC 1 and SBC 2 have equally sized clusters with 25 and 9 beams per cluster, respectively, while SBC 3 has 36 clusters of size 4, 12 clusters of size 6, and one cluster of size 9. An illustration of the spatial SINR distribution for the considered SBC schemes is available

in [4]. The results are presented using extensive Monte Carlo simulations assuming that K UEs are uniformly distributed in the network with 10^4 random realizations.

Fig. 4 shows the cumulative distribution function (CDF) of the multi-user performance for $K = 50$. It can be observed that DBC outperforms all four SBC schemes in terms of both the sum rate and fairness index, and it provides superior gains in terms of the EE performance over benchmark SBC schemes. As shown in Fig. 4(a), DBC brings about 70 Gb/s increase in the sum rate with respect to SBC 3 with 49 clusters. From the perspective of the fairness index as shown in Fig. 4(b), DBC still performs better than SBC schemes, and the performance of SBC 1 closely follows that of DBC. The reason for this is SBC 1 uses large clusters of size 25, and hence it can provide better fairness considering that the UEs are almost uniformly distributed in its clusters. The largest performance gap between DBC and SBC is manifested in EE results, as shown in Fig. 4(c), where DBC enhances the EE performance by an increase of 0.8 Gb/J over SBC schemes.

Fig. 5 shows the statistical average of the three performance metrics for both DBC and SBC for $2 \leq K \leq 100$. It can be observed that DBC outperforms SBC over the entire range of values for the number of UEs except for the average fairness index for which the performance of SBC 1 takes the lead with respect to DBC for $K \geq 64$. Interestingly, the trend of the EE performance for DBC is different from SBC, in that it is decreasing unlike the increasing trend for all SBC schemes. This is due to the fact that the numerator of the EE ratio (i.e., sum rate) is scaled with the number of clusters in both DBC and SBC, and its denominator (i.e., total PC) is fixed for SBC, while it grows linearly for DBC. However, in DBC, the rate of change in the numerator is less than the denominator, hence the decreasing trend of the EE performance.

V. CONCLUSIONS

The proposed DBC scheme effectively improves the multi-user performance in a GoB LiFi network by dynamically computing the clustering layout and providing ICI-free CoMB-JT based on the distribution of UEs. The results demonstrate the outperformance of DBC with respect to benchmark SBC schemes with fixed cluster sizes in terms of the multi-user sum rate, fairness index and EE. The use of DBC notably enhances the EE performance by 0.8 Gb/J compared with SBC schemes that use up to 49 clusters. This is of particular interest to future energy-efficient 6G networks. Nevertheless, under DBC, as the number of UEs increase, it is likely that clusters with excessively large sizes are formed, adversely affecting the performance per UE. Therefore, in our future works, we will attempt to extend the scope of the proposed DBC to address this issue and to further enhance the performance per UE.

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REFERENCES

- [1] H. Yang, A. Alphones, Z. Xiong, D. Niyato, J. Zhao, and K. Wu, “Artificial-Intelligence-Enabled Intelligent 6G Networks,” *IEEE Network*, vol. 34, no. 6, pp. 272–280, 2020.
- [2] M. R. Mahmood, M. A. Matin, P. Sarigiannidis, and S. K. Goudos, “A Comprehensive Review on Artificial Intelligence/Machine Learning Algorithms for Empowering the Future IoT Toward 6G Era,” *IEEE Access*, vol. 10, pp. 87 535–87 562, 2022.
- [3] T. Koonen *et al.*, “Ultra-high-capacity wireless communication by means of steered narrow optical beams,” *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 378, 2020.
- [4] H. Kazemi, E. Sarbazi, M. Crisp, T. E. H. El-Gorashi, J. M. H. Elmirghani, R. V. Pentty, I. H. White, M. Safari, and H. Haas, “A Novel Terabit Grid-of-Beam Optical Wireless Multi-User Access Network With Beam Clustering,” *arXiv Preprint arXiv:2404.04443 [eess.SP]*, April 2024.
- [5] C. Sun, J. Wang, X. Gao, and Z. Ding, “Networked Optical Massive MIMO Communications,” *IEEE Transactions on Wireless Communications*, vol. 19, no. 8, pp. 5575–5588, Aug. 2020.
- [6] E. Sarbazi, H. Kazemi, M. D. Soltani, M. Safari, and H. Haas, “A Tb/s Indoor Optical Wireless Access System Using VCSEL Arrays,” in *Proc. IEEE 31st Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2020, pp. 1–6.
- [7] R. Michalzik, *VCSELs: Fundamentals, Technology and Applications of Vertical-Cavity Surface-Emitting Lasers*. Springer, 2012, vol. 166.
- [8] Y. Liu, A. Wajahat, R. Chen, N. Bamiedakis, M. Crisp, I. H. White, and R. V. Pentty, “High-Capacity Optical Wireless VCSEL Array Transmitter with Uniform Coverage,” in *Free-Space Laser Communications XXXV*, vol. 12413, International Society for Optics and Photonics. SPIE, 2023.
- [9] S. Basso, H. Farooq, M. A. Imran, and A. Imran, “Coordinated Multi-Point Clustering Schemes: A Survey,” *IEEE Communications Surveys & Tutorials*, vol. 19, no. 2, pp. 743–764, 2017.
- [10] “Understanding Optical Lens Geometries,” Edmund Optics, Application Note, May 2024. [Online]. Available: <https://www.edmundoptics.com/knowledge-center/application-notes/optics/understanding-optical-lens-geometries/>
- [11] B. E. A. Saleh and M. C. Teich, *Fundamentals of Photonics*, 3rd ed. John Wiley & Sons, Inc., 2019, vol. Part I: Optics.
- [12] E. Sarbazi, H. Kazemi, M. Crisp, T. El-Gorashi, J. Elmirghani, R. V. Pentty, I. H. White, M. Safari, and H. Haas, “Design and Optimization of High-Speed Receivers for 6G Optical Wireless Networks,” *IEEE Transactions on Communications*, vol. 72, no. 2, pp. 971–990, 2024.
- [13] T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, 3rd ed. The MIT Press, 2009.
- [14] J. Carruthers and J. M. Kahn, “Angle Diversity for Nondirected Wireless Infrared Communication,” *IEEE Transactions on Communications*, vol. 48, no. 6, pp. 960–969, Jun. 2000.
- [15] S. Dimitrov, S. Sinanovic, and H. Haas, “Clipping Noise in OFDM-Based Optical Wireless Communication Systems,” *IEEE Transactions on Communications*, vol. 60, no. 4, pp. 1072–1081, April 2012.
- [16] R. K. Jain, D. M. W. Chiu, and W. R. Hawe, “A Quantitative Measure of Fairness and Discrimination for Resource Allocation in Shared Systems,” Digital Equipment Corporation, Technical Report: DEC-TR-301, 1984.
- [17] E. Björnson, J. Hoydis, and L. Sanguinetti, “Massive MIMO Networks: Spectral, Energy, and Hardware Efficiency,” *Foundations and Trends® in Signal Processing*, vol. 11, no. 3–4, pp. 154–655, 2017. [Online]. Available: <http://dx.doi.org/10.1561/20000000093>
- [18] A. Chandrakasan, S. Sheng, and R. Brodersen, “Low-Power CMOS Digital Design,” *IEEE Journal of Solid-State Circuits*, vol. 27, no. 4, pp. 473–484, 1992.
- [19] M. F. Sanya, L. Djogbe, A. Vianou, and C. Aupetit-Berthelemy, “DC-Biased Optical OFDM for IM/DD Passive Optical Network Systems,” *J. Opt. Commun. Netw.*, vol. 7, no. 4, pp. 205–214, Apr 2015.
- [20] M. T. Bohr and I. A. Young, “CMOS Scaling Trends and Beyond,” *IEEE Micro*, vol. 37, no. 6, pp. 20–29, 2017.
- [21] D. J. M. Moss, E. Nurvitadhi, J. Sim, A. Mishra, D. Marr, S. Subhaschandra, and P. H. W. Leong, “High Performance Binary Neural Networks on the Xeon+FPGA™ Platform,” in *Proc. 27th International Conference on Field Programmable Logic and Applications (FPL)*, 2017, pp. 1–4.